

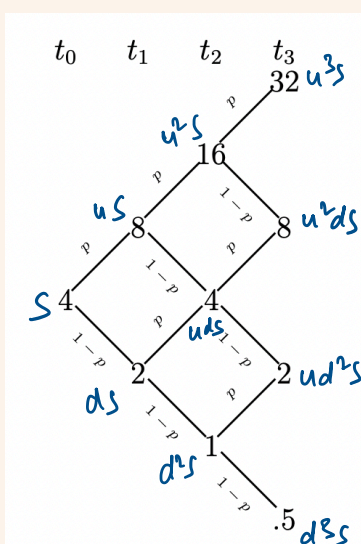
Shreejit Verma Homework 2

Thursday, 12 August 2021

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Q7

The binomial model for stock prices describes the evolution of prices in discrete time. If the price at time t is S_t , the value at time $t + 1$ will be uS_t with probability p or dS_t with probability $1 - p$. Under this model the steps are independent from one time period to the next. When $d = 1/u$ the model takes a particularly simple form and can be depicted using a tree. For example, if $S_0 = 4$, $u = 2$, and $d = 1/2$, then the model has the following structure:



In the case where $S_0 = 4, u = 2$, and $d = 1/2$, and using $p = 0.4$, find $P(S_7 \geq 32)$.

$$P(S_7 \geq 32)$$

$$\hookrightarrow 32 = 2^5 \quad \text{we have } S_0 \times u^x \times d^{7-x}$$

$$\text{we have 7 steps} \quad = 4 \times 2^x \times \left(\frac{1}{2}\right)^{7-x} \geq 32$$

$$\text{so we need minimum 5 ups}$$

$$P(S_7 \geq 32) = {}^7C_5 (p)^5 (1-p)^2 + {}^7C_6 (p)^6 (1-p) + {}^7C_7 (p)^7 (1-p)^0$$

$$= 21 (0.4)^5 (0.6)^2 + 7 (0.4)^6 (0.6) + (0.4)^7$$

$$= 0.4^5 [7 \cdot 0.6 + 1.68 + 0.16] = 0.4^5 [4.78] = 0.0962 \text{ Ans}$$

Q8

Let $X \sim \text{Geometric}(p)$ with $p < 1$. Prove that X has the memoryless property: For $n = 0, 1, \dots$ and $m = 0, 1, 2, \dots$, it holds that $P(X \geq n+m | X \geq n) = P(X \geq m)$

Soln

The probability mass function for a geometric random variable X is

$$f(x) = p(1-p)^x \quad x = 0, 1, 2, \dots$$

The probability that X is greater than or equal to x is

$$P(X \geq x) = (1-p)^x \quad x = 0, 1, 2, \dots$$

$$\text{So, } P(X \geq n+m | X \geq n) = \frac{P(X \geq n+m, X \geq n)}{P(X \geq n)}$$

$$= \frac{P(X \geq n+m)}{P(X \geq n)}$$

$$= \frac{(1-p)^{n+m}}{(1-p)^n}$$

$$= (1-p)^m$$

$$= P(X \geq m)$$

Hence, the memoryless property is proved.

Q13

The length of the life of an electronic component is adequately modelled using the exponential distribution with mean of 10000 hours. Given that the component has been working for 9000 hours, what is the probability that it will work for no more than 11000 hours? Repeat under the assumption that the lifetime (in hours) is not exponential but rather uniformly distributed over (8000, 12000).

Soln If X denotes the lifetime of the component

$$P(X > x) = \int_x^\infty \frac{1}{10000} e^{-t/10000} dt = e^{-x/10000}$$

Given that the component has been in operation for 9000 hours;

the probability that it will work for no more than 11,000 hours is

$$P(X > x | X > 9000) = \frac{P(X > 11000)}{P(X > 9000)} = \frac{e^{-11000/10000}}{e^{-9000/10000}} = e^{-(11000-9000)/10000} = e^{-2000/10000} = e^{-2/10} \text{ Ans}$$

It is the same as if it was from the start for 2000 hours.

Now, Assuming lifetime is uniformly distributed over (8000, 12000)

$$P(X \leq 11000 | X > 9000)$$

$$= \frac{P(X > 9000 \text{ and } X < 11000)}{P(X > 9000)}$$

$$= \frac{\frac{2000}{4000}}{\frac{3000}{4000}} = \frac{2}{3} \text{ Ans}$$

Q16

Suppose that $X \sim \text{exponential}(\lambda_1)$ and $Y \sim \text{exponential}(\lambda_2)$ and that X and Y are independent. What is the distribution of $\min(X, Y)$

Soln

$$E(X) = \frac{1}{\lambda_1} ; E(Y) = \frac{1}{\lambda_2}$$

$$\text{Let } Z = \min(X, Y)$$

$$X > 0, Y > 0 \Rightarrow Z > 0$$

$$f_Z(z) = 0 \text{ for } z < 0$$

Let's find the density $f_Z(z)$ of Z , for $z > 0$

$$P(Z > a) \text{ for } a > 0$$

$$P(Z > a) = P(\min(X, Y) > a) = P(X > a, Y > a)$$

$$= P(X > a)P(Y > a)$$

$$= e^{-a\lambda_1} e^{-a\lambda_2} = e^{-a(\lambda_1 + \lambda_2)}$$

$$F_Z(a) = 1 - P(Z > a) = 1 - e^{-a(\lambda_1 + \lambda_2)}$$

$$E(Z) = \frac{1}{\lambda_1 + \lambda_2}$$

$$f_Z(z) = (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2)z}$$

Z is an exponential random variable with parameter $\lambda_1 + \lambda_2$ i.e., $E(Z) = \frac{1}{\lambda_1 + \lambda_2}$

$$f_Z(z) = \begin{cases} \lambda_1 + \lambda_2 e^{-(\lambda_1 + \lambda_2)z} & \text{for } z > 0 \\ 0 & \text{for } z \leq 0 \end{cases}$$